Looking Away

Building 2: Loops

Ways between ways, and the systems that come back.

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A later chapter builds a system and blurs it by the action of an earlier one; a *loop* is the theorem saying the earlier system comes back (rule F8). Building 2 also holds the levels above moves: ways between ways (Chapter 16 is its first room).

Chapter numbers restart in this book; Building 0 and Building 1 are cited by name.

Part II runs two rules of Building 0, Chapter 0 on the map: the loop (F8) and Amplify (Z11).

Part I. Words about words

Building 0, Chapter 0 writes rules about words; Building 1, Chapter 2 builds words and glues them. Here the words are statements and proofs, and the blurs are "forget the proof" and "forget everything but the truth".

Chapter 1. Proofs

statements → what every world says · from: Building 0, Chapters 0, 3; Building 1, Chapter 2 · build, blur

Statements and proofs are words. Glue two statements when each proves the other; glue them when every world gives them the same answer. The two blurs agree: that is completeness. For the single world of the whole numbers they do not: that is incompleteness.

Written for statements built from letters with "not", "and", "or", "if". The lines for statements with "for all" are stated and reported (T4, T5); their proofs need a design of their own.

Given

G1 (Building 0, Chapter 0, M3.) Words in a list of names; maps on words by recursion.

G2 (Building 1, Chapter 2.) Words, glued by rules from below; B2 by an action on words.

G3 (Building 0, Chapter 3.) $\{0,1\}$ as $\mathbb{Z}/2$.

Definitions

D1 (Build, M3.) *Statements*: words in letters $p_1, p_2, \ldots$, the name $\bot$, and $\neg$, $\wedge$, $\vee$, $\to$. Write $A \leftrightarrow B$ for $(A \to B) \wedge (B \to A)$.

D2 (Build.) A *proof* of A: a list of statements ending in A, each one an instance of $A \to (B \to A)$, $(A \to (B \to C)) \to ((A \to B) \to (A \to C))$, $(\neg B \to \neg A) \to (A \to B)$ (with $\neg A := A \to \bot$ and $\wedge$, $\vee$ written through $\to$, $\bot$), or got from two earlier ones B and $B \to A$. Write $\vdash A$. Checking a proof is a recursion; finding one is a search.

D3 A *world*: an arrow from the letters to $\{0,1\}$. It gives every statement a value, by recursion ($\bot \mapsto 0$, and the usual tables).

The blur

On statements in the letters $p_1, \ldots, p_n$. O: $\neg$, $\wedge$, $\vee$, $\to$.

B1 (From below: forget the proof.) $A \sim B$ when $\vdash A \leftrightarrow B$.

B2 (From above: forget all but the answers.) Questions: the value of a statement in each world.

Propositions

P1 (Survives.) $\neg$, $\wedge$, $\vee$, $\to$ all respect B1 and B2. The bags of B2 form $\{0,1\}^{(\text{worlds})}$, with the operations taken world by world.

P2 (Dies: length; the proof.) p and $\neg\neg p$ are glued, of lengths 1 and 3. "Which proof" is the first thing forgotten; "how short a proof can be" dies with it, and is a question of its own (R2).

P3 (Normal form.) Each bag of B2 holds exactly one "or" of full rows, one row for each world giving 1 (in a fixed order). So there are 2^{2^n} bags.

Theorems

T1 (One way: sound.) If $A \sim B$ then A and B agree in every world. (Each axiom is 1 in every world, and "from B, $B \to A$ get A" keeps "1 in every world".)

T2 (The other way. A construction.) If A is 1 in every world, $\vdash A$. (For each world w, prove "the row of $w \to A$" by recursion on A; then remove the letters one at a time with $\vdash (p \to A) \to ((\neg p \to A) \to A)$.)

T3 (Appears: completeness.) B1 and B2 agree. So the bags of B1 are exactly 2^{2^n}, with P3 as normal forms, and "provable" is "true in every world".

T4 (Reported, classical: with "for all".) For statements about a floor with "for all" and "there is", and a list of axioms Γ: B1 (each proves the other from Γ) and B2 (same value in every floor obeying Γ) agree. The construction in the proof builds a floor out of the words themselves.

T5 (Reported, classical: Appears fails for one world.) Take the axioms of $\mathbb{N}$ (Building 0, Chapter 1, with $+$, $\times$) and replace B2 by the single world $\mathbb{N}$. Then B1 is strictly finer: some statement G is true in $\mathbb{N}$ and is not glued to $\top$. Z4 holds, Z5 fails (Building 0, Chapter 0), with G as witness. The proof codes words as numbers, so that statements about $\mathbb{N}$ speak about statements: a loop of the kind this building is named for.

Remarks

R1 (A twist.) A statement of the form "for every n, a check on n comes out yes" can be refuted by one finite witness if it is false. So if it is shown to be neither provable nor refutable, it is true in $\mathbb{N}$. Many open questions about primes (Building 0, Chapter 18) have this form.

R2 (How much proof.) Two sizes, both what dies in B1: *length*, the shortest proof, and *strength*, the first rung of a ladder of axiom lists that proves A. The pigeonhole statement for $n+1$ birds needs proofs of length growing faster than any power of n in the weakest systems (classical, reported), and only a few lines once counting bags is allowed (Building 1, Chapter II.6 uses counts this way).

R3 (Old names.) D1: propositional formulas; D2: a Hilbert system; D3: valuations; B1: the Lindenbaum algebra, a Boolean algebra; T1: soundness; T2, T3: completeness (Post; Kalmár's proof); T4: Gödel's completeness theorem; T5: Gödel's first incompleteness theorem.

R4 (Machine check, `checks/b2-ch1-2-check.py`.) The axioms on many instances are 1 in every world; $2^{2^2} = 16$ bags for two letters; P3 normal forms.

Exercises

E1 Write the normal form of P3 for $p \to q$ and for $\neg(p \leftrightarrow q)$.

E2 Show $\vdash p \to p$ from the axioms of D2 in five lines.

E3 Why does T2 not give a fast way to find proofs?

Chapter 2. Equally provable

statements → how provable they are · from: Chapter 1; Building 0, Chapters 18, 19, 34
· blur, ladder

Glue two statements when the theory proves "this one is provable exactly when that one is". Without letters the bags are a ladder of shape $\omega + 1$, and a statement's rung is the first depth at which it fails: a valuation, like v_p.

All the lines about arithmetic here rest on three classical results, reported: the conditions on $\Box$ (D1), Löb's theorem (G2), and the depth reading (G3). The chapter's own work is the blur and its reading.

Given

G1 (Chapter 1, T5.) Statements of arithmetic can speak of proofs. For a statement A, $\Box A$ is the statement "A is provable", and $\vdash A$ gives $\vdash \Box A$; $\vdash \Box(A \to B) \to (\Box A \to \Box B)$; $\vdash \Box A \to \Box\Box A$ (reported, classical).

G2 (Reported, classical: Löb.) $\vdash \Box(\Box A \to A) \to \Box A$.

G3 (Reported, classical: depth.) For statements built from $\bot$ with $\neg$, $\wedge$, $\vee$, $\to$, $\Box$ (no letters): read A at each depth $n = 0, 1, 2, \ldots$, with $\Box A$ true at depth n when A is true at every depth below n. Then $\vdash A$ exactly when A is true at every depth.

The blur

On statements of arithmetic. O: $\wedge$, $\Box$.

B1 (From below.) $A \approx B$ when $\vdash \Box A \leftrightarrow \Box B$: "equally provable".

B2 (From above, without letters.) Question: $\rho(A) :=$ the first depth at which A is false (∞ if none), the *rung* of A.

Propositions

P1 (Survives.) $\wedge$ and $\Box$. ($\Box(A \wedge B) \leftrightarrow \Box A \wedge \Box B$ by G1.)

P2 (Appears inside: Löb.) $A \approx (\Box A \to A)$ for every A. (G1 one way, G2 the other.)

P3 (Dies: truth.) $\bot \approx \neg\Box\bot$ (P2 with $A = \bot$). $\neg\Box\bot$, "consistency", is true, and $\bot$ is false.

P4 (Dies: $\vee$ and $\neg$.) $\bot \approx \neg\Box\bot$, but $\bot \vee \Box\bot$ is not provable while $\neg\Box\bot \vee \Box\bot$ is.

Theorems

T1 (Appears, without letters: the bags are the rungs.) $A \approx B \iff \rho(A) = \rho(B)$. ($\Box A$ is true at depth n exactly when $n \leq \rho(A)$; G3.)

T2 (A ladder of shape $\omega + 1$.) Every rung $k = 0, 1, 2, \ldots$ and ∞ occurs: $\rho(\Box^k\bot) = k$, and the provable statements are rung ∞. For every A with $\rho(A) = k < \infty$:

$\vdash \Box A \leftrightarrow \Box^{k+1}\bot$, so $\Box^k\bot$ is a normal form of its bag. No two rungs are glued: that would give $\vdash \Box^{k+2}\bot \to \Box^{k+1}\bot$, and Löb would give $\vdash \Box^{k+1}\bot$.

T3 (Survives, read on the rungs.) $\rho(\Box A) = \rho(A) + 1$; $\rho(A \wedge B) = \min(\rho(A), \rho(B))$. These are the laws of v_p (Building 0, Chapter 18, D4) and of ord_a (Building 0, Chapter 34, P1) with "$\times$" read as "$\wedge$": a rung is the first place a statement stops being glued to the provable ones, as $v_p(a)$ is the first rung at which a stops being glued to 0.

T4 (Placement.) $\bot$ and consistency $\neg\Box\bot$ on rung 0; $\Box\bot$ ("inconsistent") on rung 1; "$\Box\bot \to \Box\Box\bot$" on rung ∞.

Remarks

R1 (What is known.) T1–T3 are the letterless normal form of the logic of $\Box$ (Boolos, classical). Adding stronger boxes stretches the ladder to the ordinal ε_0, the measure of strength of arithmetic (Beklemishev, reported). The blur B1 with letters, and the mixture of B1 with "same consequences of the form for every n", are not worked out: open.

R2 (The twin.) The rung is a valuation: it is the fourth time the book meets "first rung at which not glued" (Building 0, Chapters 19, 21, 34). Here the ladder stops at $\omega + 1$ instead of having a bottom.

R3 (Old names.) $\Box$: the provability predicate; G1: the Hilbert–Bernays–Löb conditions; G2: Löb's theorem; G3: Solovay's completeness for GL on letterless sentences, read on a line of worlds; P3: Gödel's second incompleteness theorem (consistency is not provable) seen as "truth dies".

R4 (Machine check, `checks/b2-ch1-2-check.py`.) On letterless statements up to size 3 read at depths below 40: T1 (bags are rungs), P2 (Löb instances), T3 ($\Box$ climbs one rung, $\wedge$ is min), T4, P4.

Exercises

E1 Find the rung of $\Box\bot \to \bot$ and of $\Box\Box\bot \wedge \neg\Box\bot$.

E2 Why is $\vee$ not max on rungs? Find $\rho(A \vee B)$ for $A = \bot$, $B = \Box\bot$.

E3 Show from P2 that $A \approx \Box A \to A \approx \Box(\Box A \to A) \to (\Box A \to A)$.

Part II. The edge

Where the method stops proving and starts sorting. Two chapters state a line that is now a rule of Building 0, Chapter 0 (the loop, F8; Amplify, Z11); each chapter runs its line on the map, and says honestly what is proved, what is checked by machine, and what is reported. Nothing here is a new theorem about the primes.

Chapter 3. Loops

a later system blurred by an earlier one → what comes back · from: Building 0, Chapters 8, 16, 18, 19, 22, II.6; Building 1, Chapters II.5, II.12 · blur

Build something late, let something early act on it, and blur by the action. Sometimes the early system comes back exactly. Sometimes nearness dies and the bags cannot be told apart: then keep the moves. One prime alone is tame; two primes together are already wild.

The line of this chapter

L (Loop; rule F8 of Building 0, Chapter 0.) A chapter K builds a system X and blurs it by the action of an earlier system S (an orbit blur, Building 1, Chapter II.5). A *loop* is a theorem of K that S comes back: a question on the bags whose answers match S, respecting the operations named. Verdicts: *closed* (S comes back exactly); *changed* (larger or smaller); *wild* (the bags are not kept apart by any rung of nearness: keep the moves, do not forget them). No chapter before K may use the loop (F7).

Propositions: the loops on the map

P1 (Closed: $\mathbb{Z}$ on $\mathbb{R}$.) Blur $\mathbb{R}$ by $\mathbb{Z}$ acting by $+$ (Building 0, Chapter 8). The turns of loops in $\mathbb{R}/\mathbb{Z}$ are whole numbers, and every whole number occurs (Building 0, Chapter 16, T6): $\mathbb{Z}$ comes back.

P2 (Tame: one prime.) Let the powers of p act on $(0,\infty)$ by $\times$, and read through log (Building 0, Chapter 22): the bags are a circle of length $\log p$, a picture of Building 0, Chapter 8. Likewise the units of $\mathbb{Z}_p$ acting on $\mathbb{Z}/p^k$ by $\times$ have exactly $k+1$ bags, one per valuation (Building 0, Chapter 19).

P3 (Wild: two primes.) Let 2 and 3 act on $(0,\infty)$ by $\times$. Read through log, the orbit of 1 is $\{a\log 2 + b\log 3\}$, and $\log 3/\log 2$ is not a fraction ($2^a = 3^b$ has no answer with $a, b \geq 1$, by unique factorisation). So the orbit is dense (Building 0, Chapter 8, T7, irrational turns): every rung of nearness glues every bag to every other. Nearness dies; the loop is wild. Machine: the smallest gap among $a\log 2 + b\log 3 \leq L$ falls as L grows ($L = 10, 20, 40, 80$).

P4 (Closed, with the moves kept: $\mathbb{Z}/p$.) On $\mathbb{Z}/p$, $+$ gives the waves $\psi(x) = e^{2\pi i x/p}$ and $\times$ gives the characters χ of the clock $(\mathbb{Z}/p)^\times$ (Building 1, Chapter II.12, T5). Mix them: $\tau(\chi) := \sum_x \chi(x)\psi(x)$. Then for every $\chi \neq 1$

$$|\tau(\chi)|^2 = p.$$

(Write $|\tau|^2 = \sum_{x,y} \chi(x)\overline{\chi(y)}\psi(x-y)$; put $x = ty$ and sum over y first: the sum of a wave over a clock is $p-1$ or -1; then $\sum_t \chi(t) = 0$.) A size comes free from $+$ (Parseval) and χ respecting $\times$.

P5 (Lands on a known line: det acting on "divides".) The $n \times n$ matrix R with $R_{ij} = 1$ when $j = 1$ or $i \mid j$ has $\det R = M(n) = \sum_{k \le n} \mu(k)$ (classical, Redheffer; checked $n \le 40$). The line $|M(n)| \le C_\varepsilon n^{1/2+\varepsilon}$ is equivalent to the Riemann hypothesis (classical, reported).

Theorems

T1 (What comes back, sorted.) On the map: loops by $+$ give a mirror (P1; on every closing-up of $\mathbb{Q}$ at once, $\mathbb{Q}$ comes back as its own waves: reported, Tate); loops by $\times$ give the primes (P2; reported, class field theory for $\mathbb{Q}$); the loop mixing them, $\mathbb{Q}^\times$ acting on all closings-up with $+$, is wild (P3 is its smallest piece: reported, Connes).

T2 (Wildness is not local.) Every single prime gives a tame loop (P2). Two primes already give a wild one (P3). So no object built one prime at a time can carry a size on the mixed loop.

Remarks

R1 (What the method did.) The loop line sorts what is on the map: closed, tame, wild. It proves P1–P4 and T2; T1's three verdicts are known mathematics, placed.

R2 (Finite floors, moves kept.) On curves over $\mathbb{Z}/p$ every count's error is a sum of mixed sums like P4, each of size $\sqrt{p}$ (Jacobi sums; checked for $x^3 + y^3 = 1$, 20 primes). On $\mathbb{Z}$ the same mixed sums appear in the mirror of a series, not in its zeros.

R3 (Old names.) P4: Gauss sums; T1: Tate's thesis (self-duality of the adeles), class field theory, the adele class space; P5: the Redheffer matrix, the Mertens function.

R4 (Machine check, `checks/b2-p2-check.py`.) P4 for every character mod every prime $p < 80$; P2, P3 gaps; P2's $k+1$ bags for 2^5, 3^4, 5^3; P5 for $n \le 40$.

Exercises

E1 Compute $\tau(\chi)$ for the sign character mod 5 ($\chi(x) = \pm 1$ by squares).

E2 Why is the loop of 2 and 4 acting by $\times$ tame?

Chapter 4. Amplify

a bound on every rung → a sharp bound · from: Building 0, Chapters 5, 17, 20, 28, 34; Building 1, Chapters II.12, III.3 · ladder, blur

Blur, moves and loops ask "same or different?". Amplify asks "how much?". Climb the powers of one move, bound every rung with any constant, and blur sequences by growth: the constant dies and the bound becomes sharp.

The line of this chapter

L (Amplify; rule Z11 of Building 0, Chapter 0, drafted as Z12.) Given a move, its power ladder, and rung counts $S_m = \sum_i c_i \alpha_i^m$ with finitely many distinct α_i and every $c_i \neq 0$. Blur sequences by growth: $a \preceq b$ when $\limsup |a_m|^{1/m} \leq \limsup |b_m|^{1/m}$ (one-way; Building 0, Chapter 17).
Survives: "the largest $|\alpha_i|$". So if $|S_m| \leq C(m+1)^d \rho^m$ for every m, with any C and d, then every $|\alpha_i| \leq \rho$.
Dies: the constant C, and $(m+1)^d$ ($C\rho^m$ and ρ^m are glued).

T1 (Proof.) The bound makes $\sum_m S_m t^m$ converge for $|t| < 1/\rho$ (Building 0, Chapter 20, comparison). That sum is $\sum_i c_i \alpha_i t/(1-\alpha_i t)$, whose poles are exactly the $1/\alpha_i$ (distinct, no $c_i = 0$); a series that is a fraction of polynomials converges exactly out to its nearest pole (Building 0, Chapter 34, T1 and Chapter 33, P2). So no $1/\alpha_i$ is inside $|t| < 1/\rho$.

Propositions: three examples

P1 (Cauchy–Schwarz with constant 1.) Suppose only that $|\langle x, y\rangle| \leq C|x||y|$ for all lists of all lengths, one C for all. Apply it to $a^{\otimes k}$, $b^{\otimes k}$ (Building 1, Chapter II.8, T3): $\langle a^{\otimes k}, b^{\otimes k}\rangle = \langle a, b\rangle^k$ and $|a^{\otimes k}| = |a|^k$, so $|\langle a, b\rangle| \leq C^{1/k}|a||b|$ for every k, and $C^{1/k}$ comes near 1. Dies when the constant depends on the length: n^k on rung k gives back n.

P2 (The trace finds the largest eigenvalue.) For a square matrix A with eigenvalues λ_i, $\operatorname{tr} A^m = \sum \lambda_i^m$ (each with its multiplicity, never 0), and $\lim |\operatorname{tr} A^m|^{1/m}$ along a suitable part is $\max |\lambda_i|$ (T1; one entry of A^m can hide an eigenvalue, the trace cannot).

P3 (The road to Building 1, Chapter III.3, T3.) Move: $x \mapsto x^p$ on a curve over $\mathbb{F}_{p^m}$ (Building 1, Chapter II.12). Rung m: $S_m = p^m + 1 - N_{p^m}$. If $S_m = \alpha_1^m + \cdots + \alpha_{2g}^m$ with $\alpha_i \alpha_{i+g} = p$ (reported: rationality and the mirror) and $|S_m| \leq C p^{m/2}$ for every m (reported: a rough bound by an auxiliary polynomial), then $|\alpha_i| \leq \sqrt{p}$ by L, and $= \sqrt{p}$ by the mirror; with $g = 1$, $|N_p - p - 1| \leq 2\sqrt{p}$. Two inputs reported; Amplify itself is proved.

Theorems

T2 (What Amplify never does.) It removes losses that do not grow with the rung. A loss of a fixed factor per rung (C^m) is not removed: it changes ρ.

T3 (On $\mathbb{Z}$, one slot is empty.) For the zeros of the series ζ there are power ladders with a Close step (reported: Li's ladder, $s \mapsto 1 - 1/s$; the cut ladder $x \mapsto x^m$ in

$\psi(x) - x$; convolution powers of a test map). What no one has is the weak bound: a bound on every rung losing only a constant. Every known bound loses a fixed factor per rung (T2).

Remarks

R1 (What the method did.) Amplify is built from parts already in the books: a move (Building 0, Chapter 0, W2), a ladder (Chapter 5), the growth blur (Chapter 17), and a Close step. Its proof is T1. The three proofs of the curve line (Stepanov–Bombieri, Weil, Deligne) are one chain read this way. Known mathematics (amplification, the tensor power trick); its place in the grammar is the book's.

R2 (Finite floors, checked.) The genus-2 curve $y^2 = x^5 + 2x + 1$ over $\mathbb{F}_{3^m}$: $N = 7, 15, 19, 83$ for $m = 1$ to 4; the four α_i from N_1, N_2 all have size $\sqrt{3}$ and predict N_3, N_4 exactly. Graphs: a ladder whose weak bound fails is not saved; the bound is the whole content.

R3 (Old names.) L: amplification; P1: Cauchy–Schwarz by tensor powers; P2: Gelfand's formula (trace form); P3: Hasse's theorem by Weil's route; T3: Li's criterion (Li; Bombieri–Lagarias).

R4 (Machine check, `checks/b2-p2-check.py`.) P1's identities for $k \leq 4$; P2 on a random 5×5 matrix; R2's counts in $\mathbb{F}_{3^m}$, $m \leq 4$, and the predictions.

Exercises

E1 Use P1's route to show $|\sum a_i b_i c_i| \leq (\sum a_i^3)^{1/3} (\sum b_i^3)^{1/3} (\sum c_i^3)^{1/3}$ for $a_i, b_i, c_i \geq 0$, given it with some constant.

E2 Why does P2 fail for one entry of A^m? Find a 2×2 example.

Chapter 5. Atoms

things built by a glue → their bags of atoms · from: Building 0, Chapter 18; Building 1, Chapters II.3, II.7, II.14 · blur, ladder

Forget how a thing was glued from atoms and keep only the bag of atoms. Sometimes the bag tells everything, sometimes only its size, sometimes nothing. And a floor with two glues is where the hard lines live.

The blur

On a floor with one glue (an operation) whose members are built from atoms (members that cannot be glued from smaller ones). B1: forget the order and the brackets of the gluing; keep the bag of atoms.

Propositions: a ladder of floors

P1 (Rung 1, free: one bag builds one thing.) $\mathbb{Z}$ under $\times$ (Building 0, Chapter 18, T3); $K[x]$ (Building 1, Chapter II.7, T2); $\mathbb{Z}[i]$.

P2 (Rung 2, fair: the bag of rungs survives, the glue does not.) Finite move systems, glued by ladders of blurs: the list of simple rungs is kept (Building 1, Chapter II.14, T3), but $\mathbb{Z}/4$ and $\mathbb{Z}/2 \times \mathbb{Z}/2$ have the same bag. $\mathbb{Z}[\sqrt{-5}]$: factorisations differ (Building 1, Chapter II.7, P3) but every factorisation of a member has the same number of atoms (classical, Carlitz; checked to size 600).

P3 (Rung 3, unfair: not even the number survives.) $\mathbb{Z}[\sqrt{-14}]$: 3 and $5 \pm 2\sqrt{-14}$ are atoms (no member has size 3 or 27), and

$$81 = 3 \cdot 3 \cdot 3 \cdot 3 = (5 + 2\sqrt{-14})(5 - 2\sqrt{-14}),$$

four atoms against two.

P4 (Rung 4, atomless.) $\mathbb{Z}$ under $+$: $\mathbb{Z} \supset 2\mathbb{Z} \supset 4\mathbb{Z} \supset \cdots$ never ends; $\mathbb{Q}$, $\mathbb{R}$. Use generators or nearness instead.

P5 (Rung 5, uncomputable.) Move systems given by words and gluings: whether two words are glued cannot be decided in general (Building 1, Chapter 2, T4, reported).

Floors with two glues

P6 (The table.)
$\mathbb{Z}$: free under $\times$, atomless under $+$; $+$ does not respect the primes; the line about primes (Chapter 6) open.
Polynomials over $\mathbb{Z}/p$: free under $\times$; $+$ has atoms ($p \cdot 1 = 0$); irreducibles counted exactly (Building 1, Chapter II.12, T8).
Curves over $\mathbb{Z}/p$: fair under $\times$; $+$ has atoms; a move ($x \mapsto x^p$) respects both glues; the line proved (Chapter 4, P3).
$(0, \infty)$: atomless under both, and log turns one into the other: nothing to prove.
Atoms with sizes but no exact $+$ (Beurling's primes): the line can fail (reported).

P7 (Observation, not a theorem.) The line is proved where $+$ has atoms and one move respects both glues; it is open where $\times$ is free and $+$ is atomless; with no exact $+$ it can fail.

Remarks

R1 (What the method did.) One blur (forget the glue) and a ladder of floors by how much it keeps. P7 restates the known gap between characteristic p and characteristic 0 (the "field with one element" programme) as two columns of one ladder.

R2 (Old names.) P1: unique factorisation; P2: half-factorial domains (class number 2, Carlitz), Jordan–Hölder; P3: elasticity; P5: the word problem.

R3 (Machine check, `checks/b2-p2-check.py`.) Lengths of all factorisations in $\mathbb{Z}[i]$ and $\mathbb{Z}[\sqrt{-5}]$ to size 600 (always equal); P3.

Chapter 6. The missing move

the primes → a Survives line with no floor · from: Chapters 3–5; Building 0, Chapters 17, 18, 34 · survey

The Riemann hypothesis written as a line of the grammar: a Survives line for which no floor has been built. Each attempt in this book moves the missing piece somewhere else, and every road ends at one place: an exact cancellation.

The line

L (Wanted.) A floor X, built from $\mathbb{Z}$, with a move ϕ respecting both $+$ and $\times$, whose power ladder has a weak bound (Chapter 4, L), and an arrow back to the primes under which the eigenvalues of ϕ are the zeros of ζ. Then Amplify gives the line. Nobody has X.

Propositions: what was found, with labels

P1 (Where the line sits, reported.) The line is equivalent to: Weil's quadratic form in the primes is ≥ 0 on every test map (classical). Cut at size λ (only $n \leq \lambda^2$ used), the form's lowest test map has its zeros within 10^{-3} of the first eight zeros of ζ already at $\lambda^2 = 10$ (checked; known: Connes–Consani).

P2 (Where a bad zero hides; checked, novelty unchecked.) Plant one zero off the line at height T: the smallest cut at which the form turns negative is $\lambda^2 \approx T/2\pi$, almost independent of the distance from the line. A bad zero hides by height, not by nearness.

P3 (Every prime is needed; checked.) Remove one prime p well inside the cut and the form drops to $-\log p/\sqrt{p}$; keep only finitely many primes and no size appears at all. The size is global (matches Chapter 3, T2).

P4 (A finite floor that carries the line; checked to $n = 300$, probably known.) The characteristic polynomial of the matrix R of Chapter 3, P5 is $(x-1)^{n-K-1}Q_n(x-1)$ with $K = \lfloor \log_2 n \rfloor$ and

$$Q_n(y) = y^{K+1} - \sum_{k=1}^{K} F_k(n)\, y^{K-k},$$

$F_k(n)$ the number of chains $1 < a_1 \mid a_2 \mid \cdots \mid a_k \leq n$ with proper steps: glue counts of $\times$. The line is equivalent to: the small roots have geometric mean coming near $1/\sqrt{2}$ at most. Random sequences with the same shape fail: it is a cancellation, not a shape.

P5 (The mixed condition; checked.) A series with a mirror but no prime product (Davenport–Heilbronn) fails the cut form at $\lambda^2 \approx 40$, caught by its zero off the line; turning its extra weights down does not repair it. Positivity needs the prime data and the mirror to belong to one object.

Theorems

T1 (Every road ends at cancellation.) In Building 0, Chapter 17 the orders-of-magnitude blur kills cancellation (its P6). In Chapter 4 the empty slot is a bound that loses

only a constant, which needs cancellation. In P4 the condition is an alternating sum of glue counts. In P2, P3 the size is exact and global. So the missing piece is one thing seen four ways: an exact cancellation among all primes at once.

Remarks

R1 (Honest state.) Nothing here proves anything new about ζ. The books sort the known proofs into one chain (Chapter 4), name the empty slot, and give three checked numerical facts (P2, P3, P4) whose novelty has not been checked against the literature.

R2 (Corrections kept on record.) "$\mathbb{Z}$ is an island" (no nearby prime data passes) was too strong: the passing set is convex and other weights pass at every finite cut. "The primes are the top point of the passing set" holds only for the first few primes ($2, 3, 5, 7, 11, 13, 19$ at $\lambda^2 \leq 100$).

R3 (Old names.) L: a Hilbert–Pólya operator, the cohomology of $\operatorname{Spec} \mathbb{Z}$; P1: Weil's explicit formula and positivity criterion; P5: the Davenport–Heilbronn function, the Selberg class.

R4 (Sources and checks.) The files `LOOP-RULE.md` and `amplify/` in the folder `riemann` of the project (scripts and outputs dated 11–12 Nov). P4's formula is checked in `checks/b2-p2-check.py` for $n = 6, 12, 30, 48$.